Discrepancy Investigation 2024-09-28

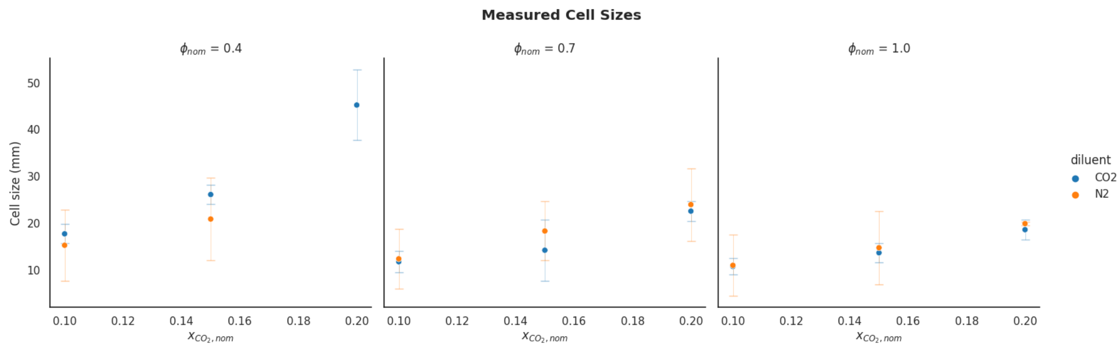
September 28, 2024

1 Measurements

1.1 Cell Size

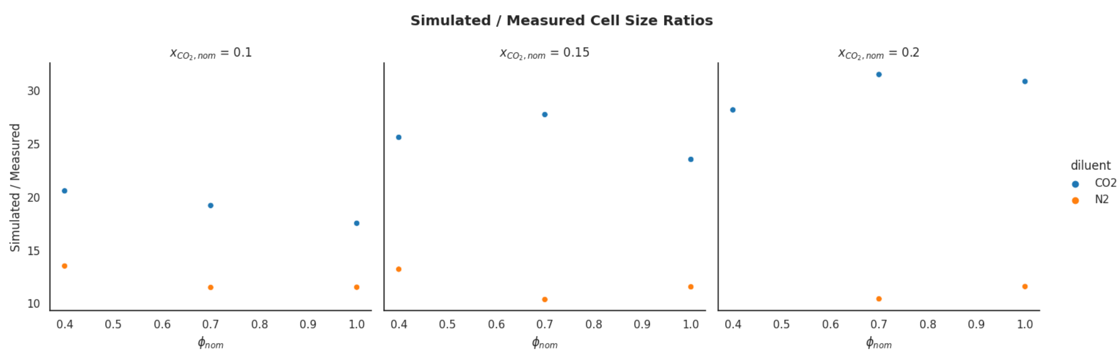
1.1.1 Measured Cell Sizes

See section 3.2-3.3 of preliminary proposal for a description of measurement methods



1.1.2 Cell Size Measurements vs. Simulations

See section 3.4 of preliminary proposal for a description of simulation methods

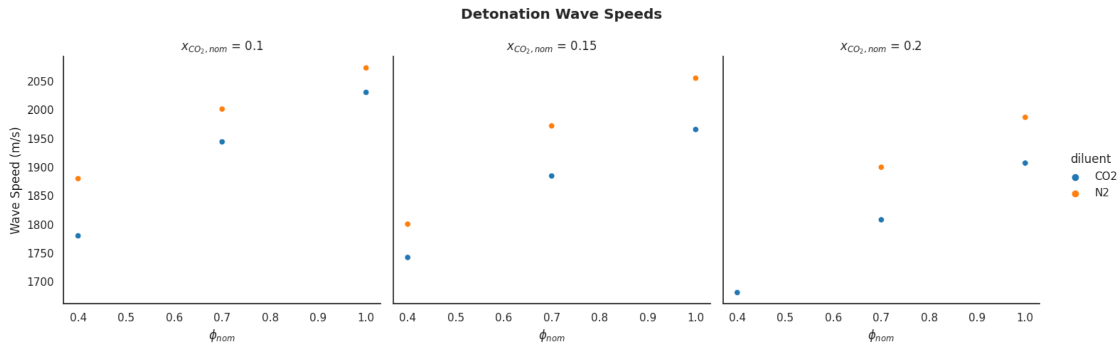


We see a divergence between N2 and CO2 diluted measurements and simulations of cell sizes corresponding to CO2 mole fraction

1.2 Wave Speed

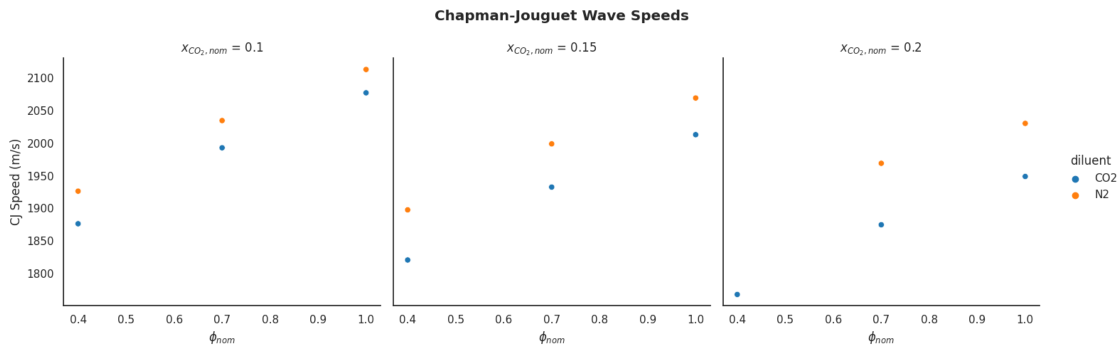
1.2.1 Measured Wave Speed

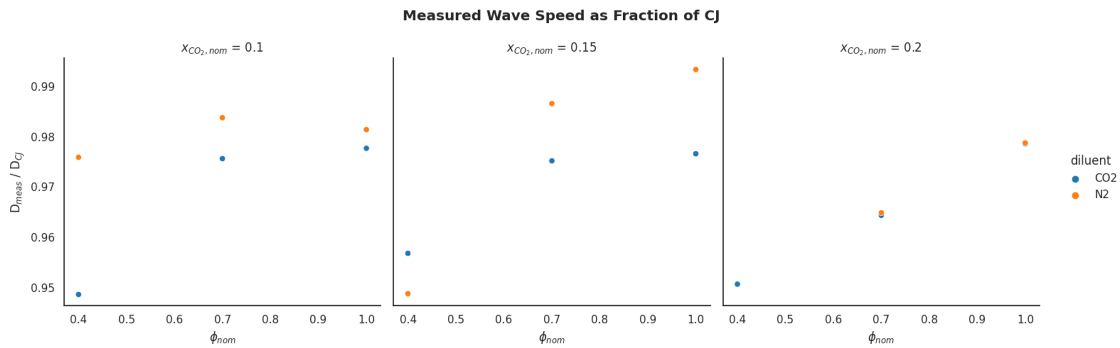
See section 3.2-3.3 of preliminary proposal for a description of measurement methods



1.2.2 CJ Speed

See section 3.4 of preliminary proposal for a description of CJ speed calculation methods





2 Potential Causes of Divergence

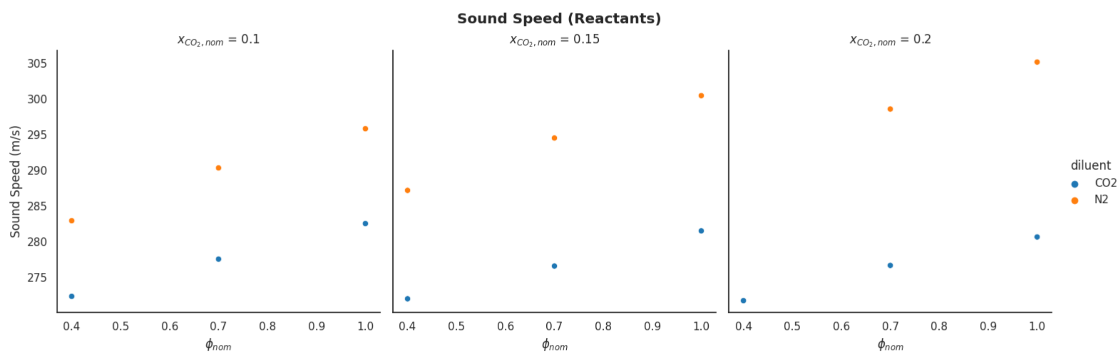
2.1 Sound Speed

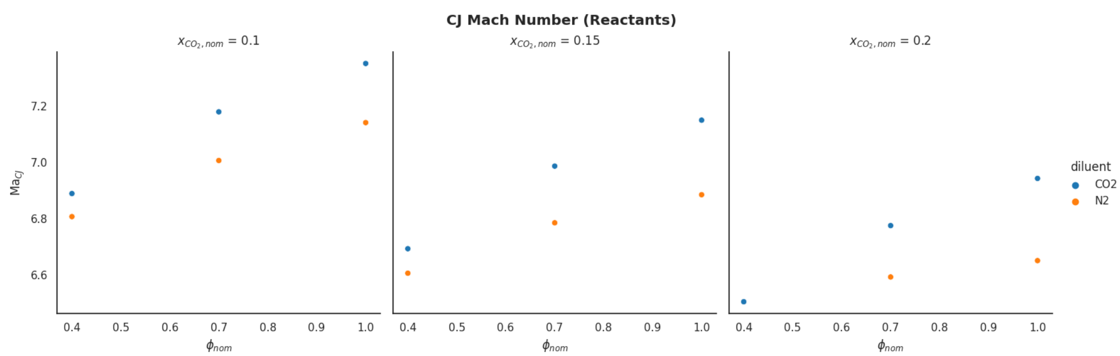
2.1.1 Methodology

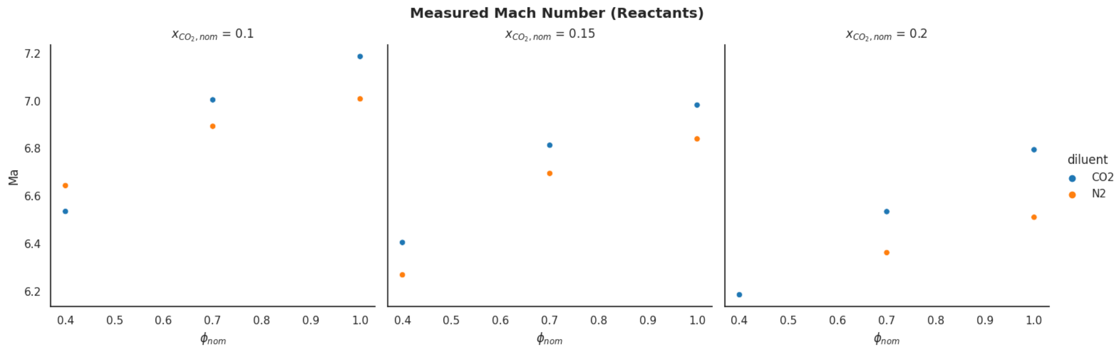
Sound speed was calculated using the `sdtoolbox.thermo.soundspeed_fr()` function. This function calculates the sound speed in a frozen reactant or product mixture (i.e. no reactions are allowed to take place during the calculation). `soundspeed_fr` uses the following methodology:

1. The density, pressure, and mass entropy of the gas mixture are collected
$$\rho_0, P_0, S$$
2. The mixture's density is perturbed to 100.1% of its original value while the entropy is held constant
$$\rho_1 = 1.001\rho_0$$
3. The perturbed density and corresponding pressure are used to calculate sensitivity of pressure to density at constant entropy
$$\left.\frac{dP}{d\rho}\right|_S = \frac{P_1 - P_0}{\rho_1 - \rho_0}$$
4. The sound speed is calculated
$$a = \sqrt{\left.\frac{dP}{d\rho}\right|_S}$$

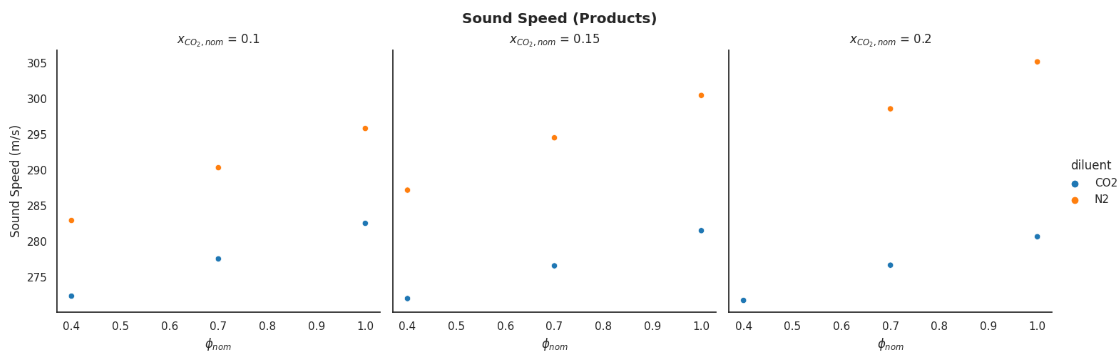
2.1.2 Reactants

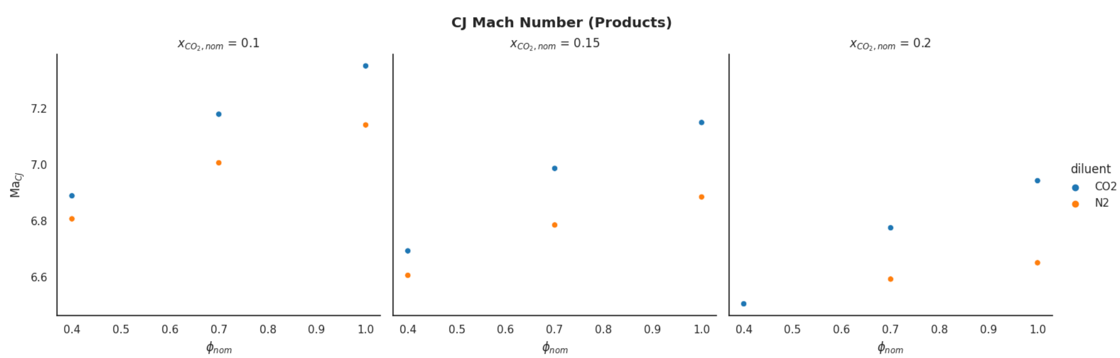


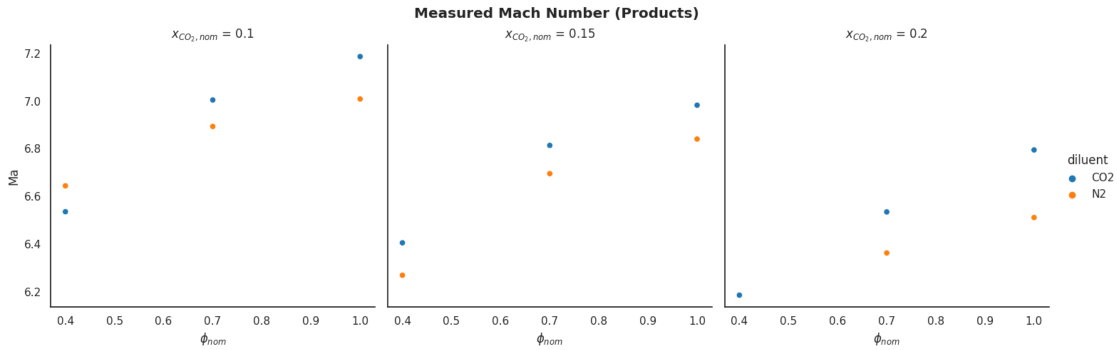




2.1.3 Products







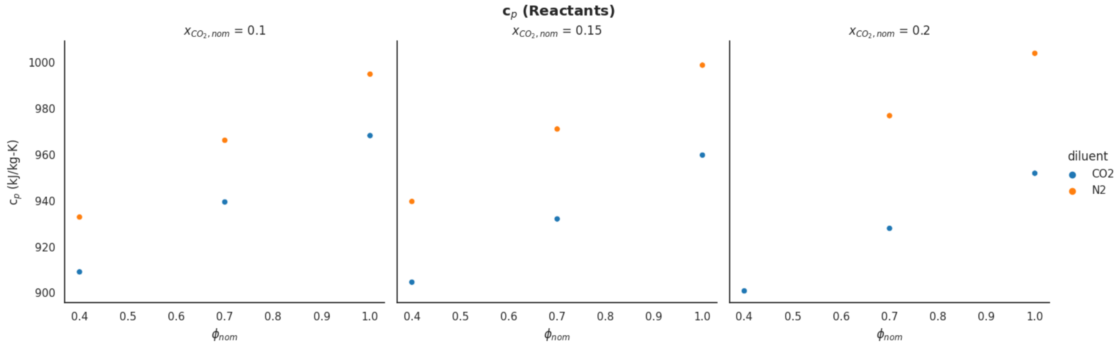
2.2 Specific Heat Ratio

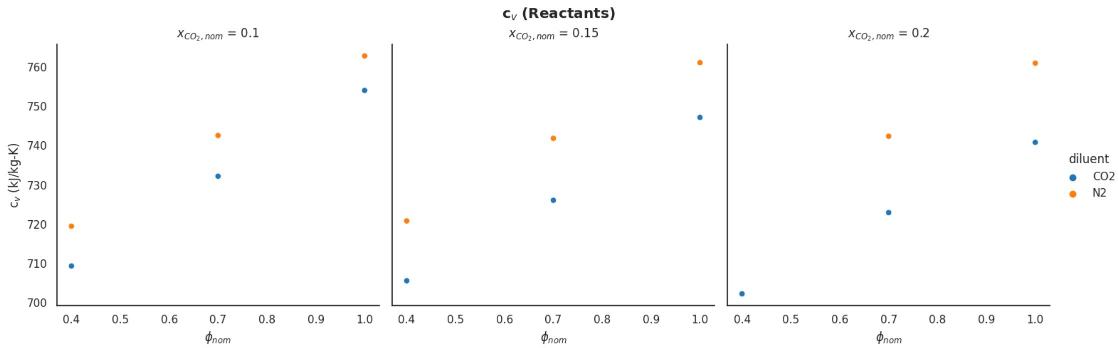
2.2.1 Methodology

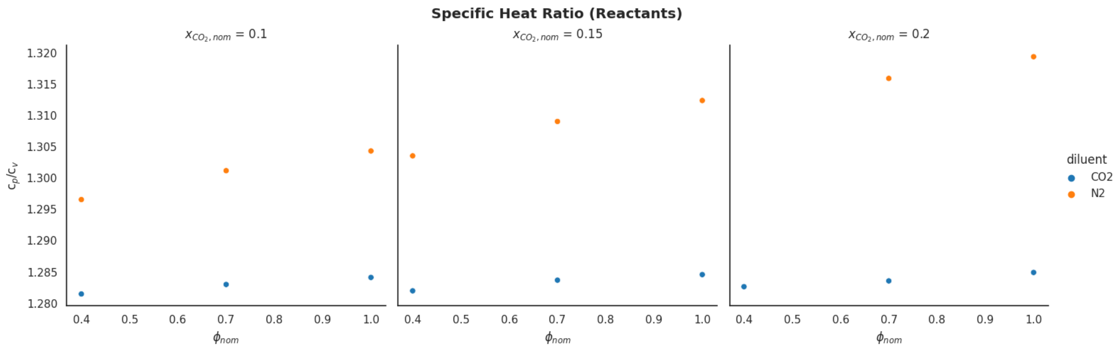
Specific heat ratio is calculated using the physical properties from a `cantera.Solution` object at the state of interest

$$\frac{c_p}{c_v}$$

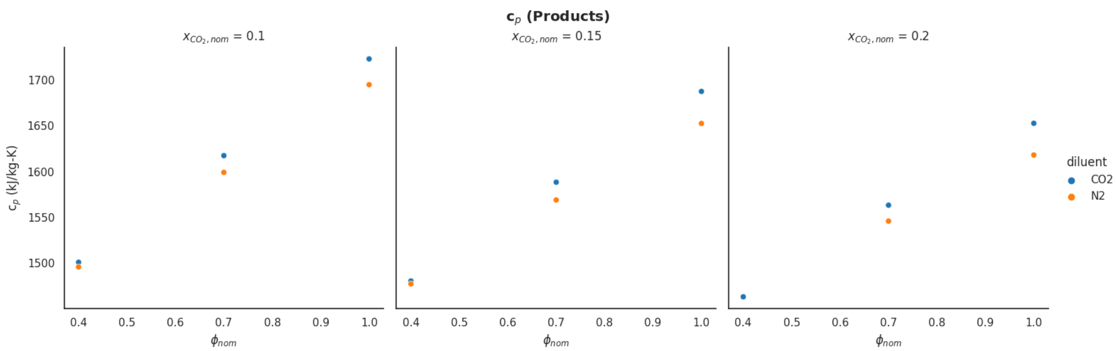
2.2.2 Reactants

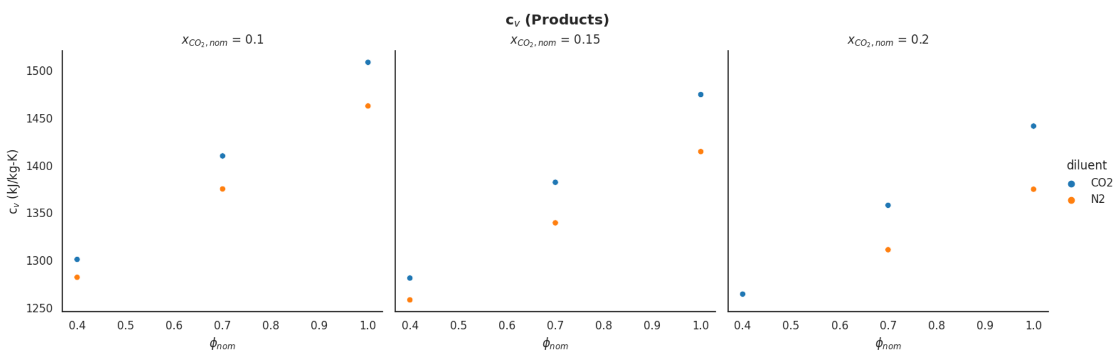


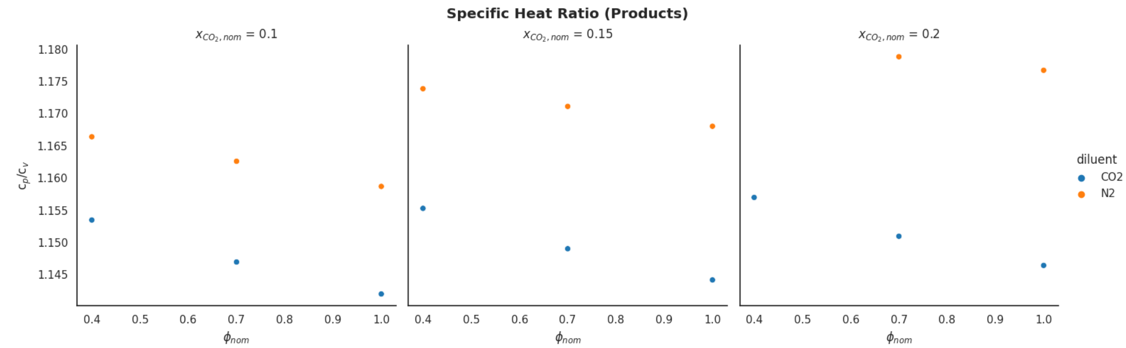




2.2.3 Products



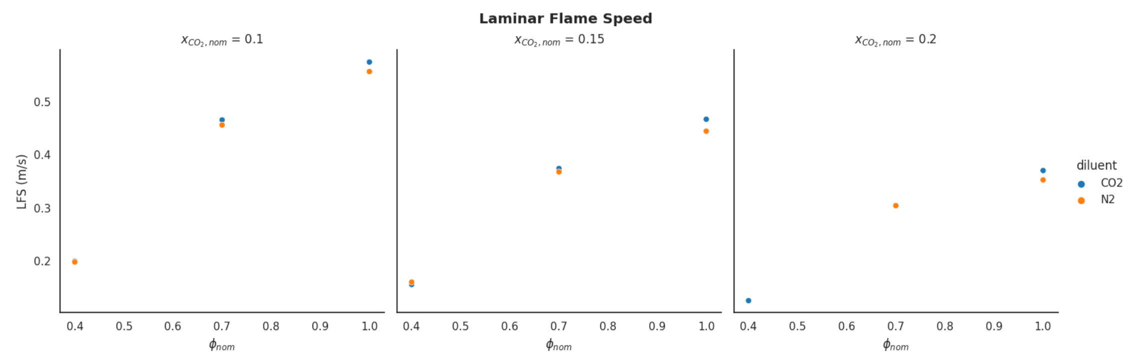


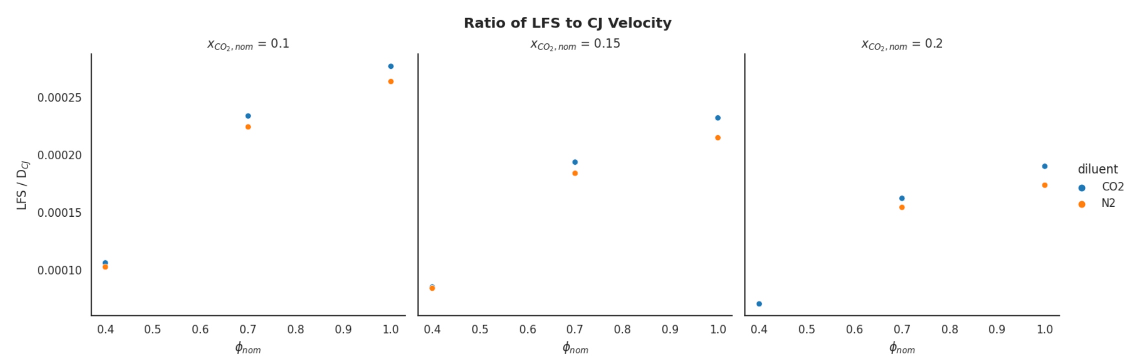


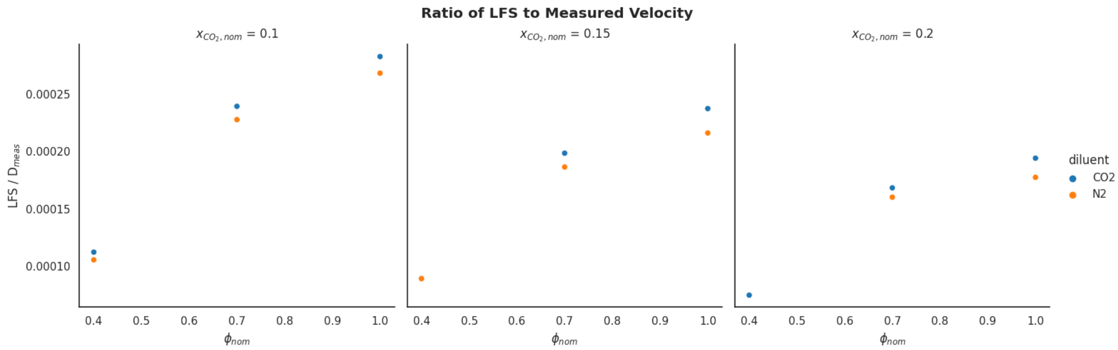
2.3 Laminar Flame Speed

2.3.1 Methodology

Laminar flame speed is read from the first velocity value calculated using `cantera.FreeFlame` with refine criteria of `ratio=3, slope=0.1, curve=0.1`







Ratio of LFS to Measured Velocity
$x_{CO_2,nom}$ = 0.1
$x_{CO_2,nom}$ = 0.15
$x_{CO_2,nom}$ = 0.2
LFS / D_{meas}
0.00025
0.00020
0.00015
0.00010
0.4 0.5 0.6 0.7 0.8 0.9 1.0
ϕ_{nom}
diluent
CO2
N2